mmdb - ex07

1.1 CBR = Content based retrieval is a search / query where objects are retreived by their extracted content (color, shape, texture).

1.2 CBR Features:

- Feature extractor: image processing algorithms that extract numerical descriptors like color histogram
- Indexer: high-dimensional indexing structures like Kd tree, VP-trees
- Query Processor = Parses the query, computes its result, runs similarity search
- Similarity = Computes how close two feature vectors are
- Ranking = Returns top-K matches ordered by distance

1.3 Feature vector is a numerical representation of a multimedia object

1.4: Subject of similarity = a gap between what humans percieve and what computers percieve

- Semantic gap = low-level features doesn't translate to high level features
- User intent is hard to capture = a query image might be an example of colour, composition, subject, or style
- Relevance feedback is needed = the user must refine the query
- Curse of dimensionality = feature are often high-dimensional, which makes neares neighbour became inefficient

2. - Dominant Color = Few colors that occupy the majority of an image (MPEG-7)
- Spatial coherency = in CBR visual features are not just present, but are arranged in a consistent spatial way (view images always has sky above)
- Distance metrics = A math rule that answers the amount of the similarity between a database item to a query (euclidean metrics)
- Curse of dimensionality = when vectors have many dimensions (as often happen), distance based similarity became inefficient (nearest-neighbor)
- Types of content based queries:
  - Query by example = Sample audio
  - Query by sketch = draws rough shape
  - -11- Feature specs = set of range (eg ≥ 30% blue)
  - -11- Concept = Set high-level concept (eg: a man doing sports)
  - -11- Relevance feedback = User returns result as relevant / irrelevant

3. Conditions = • the images must have a truck within it
• the truck must be predominantly blue

problems = • how much "blue" is does the user accept. Some may be okay if a truck is only 30% blue as long as blue is the dominant color or some may only accepts it if the truck is > 50% blue
• a "truck" might also be an "SUV" which is classified as a car but is also sometimes be called as a truck

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Bin	Colour
0	(0,0,0) Black
1	(0,0,1) blue
2	(0,1,0) green
3	(0,1,1) cyan
4	(1,0,0) red
5	(1,0,1) magenta
6	(1,1,0) yellow
7	(1,1,1) white

quantization for 8 colors

left image
8 pixels in bin 1
8 pixels in bin 7

right image
8 pixels in bin 0
8 pixels in bin 7

histogram

Image	black	blue	green	Cyan	red	magenta	Yellow	White
left	0	8	0	0	0	0	0	8
right	8	0	0	0	0	0	0	8

even bin quantizations for 2 bits (4 bins)
Black, blue = Bin 0 = no green no red
green, cyan = Bin 1 = no red
red, magenta = Bin 2 = no green
Yellow, white = Bin 3 = green, red

Images	B_0	B_1	B_2	B_3
left	8	0	0	8
right	8	0	0	8

with 2-bits quantization the images became indistinguishable

5.1 Minkowski Distance

$\hookrightarrow L_1 = |0-8| + |8-0| + |0-0| + |0-0| + |0-0| + |0-0| + |0-0| + |8-8| = 8+8 = 16$

$L_2 = \sqrt{(0-8)^2 + (8-0)^2 \cdots\cdots + |8-8|^2} = \sqrt{64+64} = \sqrt{128}$

$L_\infty = \max_i |X_i - Y_i| = 8$

5.2 Statistical difference: $H_1 = (5,5,5,5)$ $H_2 = (8,5,4,3)$

10 15 20   8 13 17 20

Cum sum$(H_1) = (5, 10, 15, 20)$

Cum sum$(H_2) = (8, 13, 17, 20)$

$\text{Diff} = (|5-8|, |10-13|, |15-17|, |20-20|)$
$= (3, 3, 2, 0)$

$K_s \max = 3$

Chi squared distance

$\text{Bin}_1 = (5-8)^2 / (5+8) = 9/13 = 0,692$

$\text{Bin}_2 = (5-5)^2 / (5+5) = 0$

$\text{Bin}_3 = (5-4)^2 / (5+4) = 1/9 = 0,111$

$\text{Bin}_4 = (5-3)^2 / (5+3) = 4/8 = 1/2 = 0.5$

$X^2 = 0,692 + 0 + 0,111 + 0,5 = 1,303$